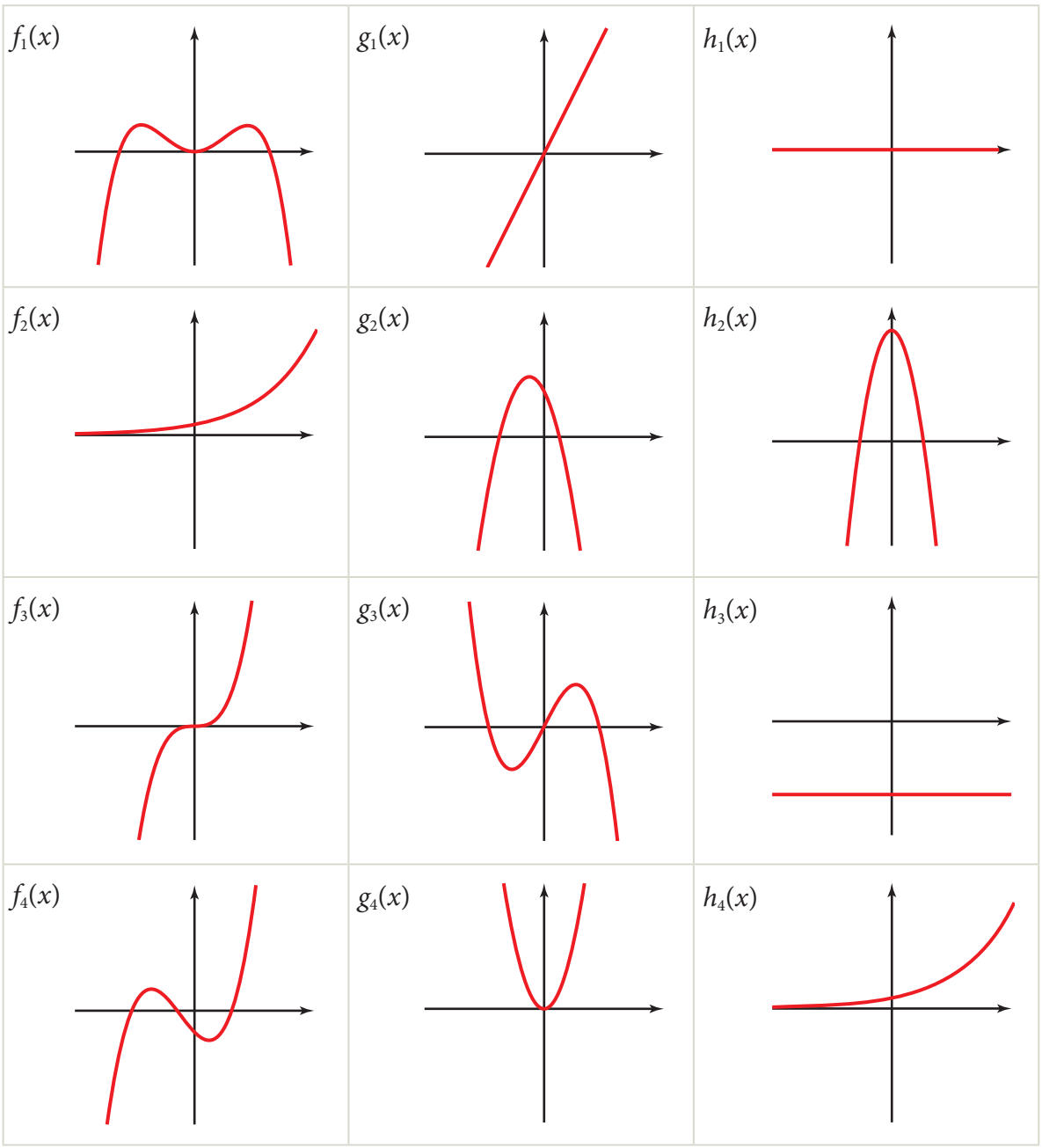
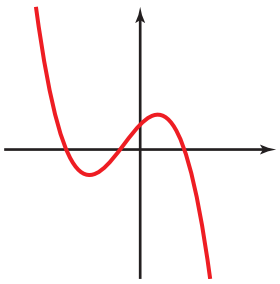
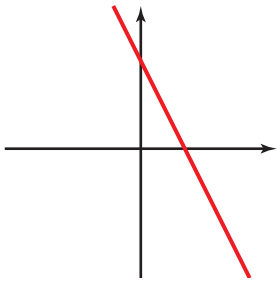
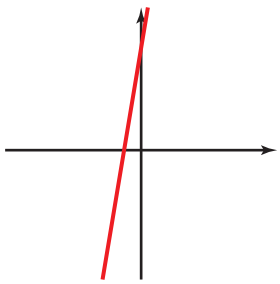
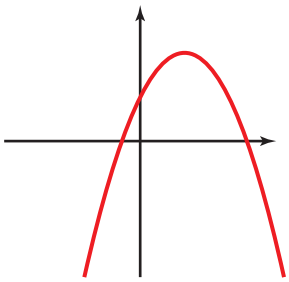
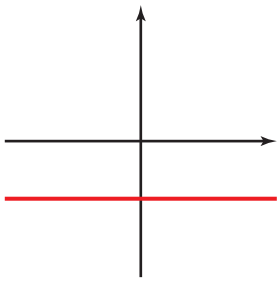
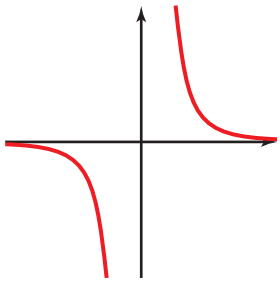
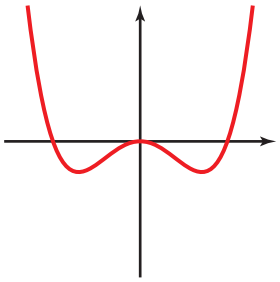
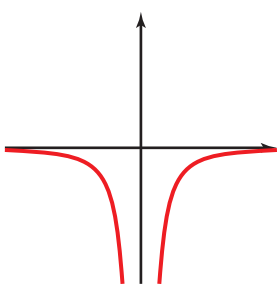
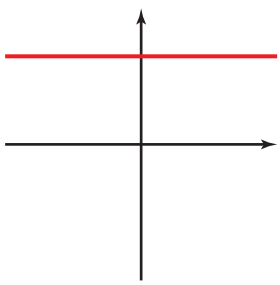
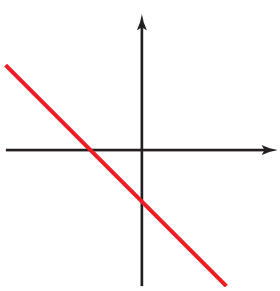
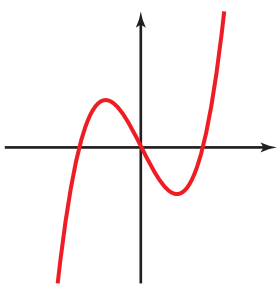
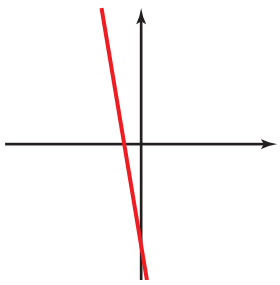
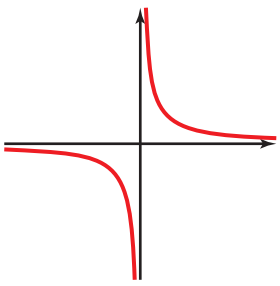
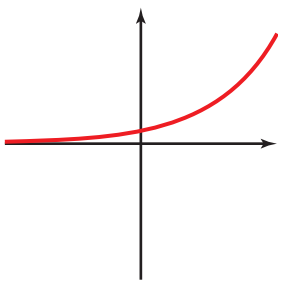
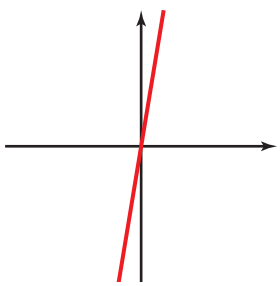
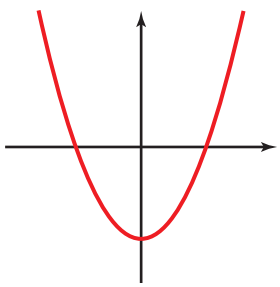
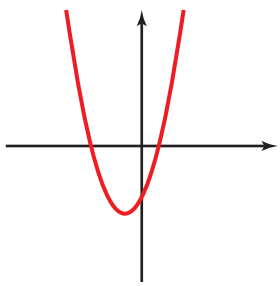
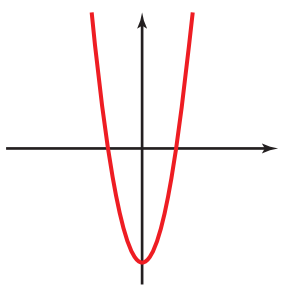
WORKSHEET

Curve sketching with derivatives

1 The graphs of 10 functions $f_1(x)$ to $f_{10}(x)$, 10 gradient functions $g_1(x)$ to $g_{10}(x)$ and 10 second derivative functions $h_1(x)$ to $h_{10}(x)$ are shown below. Match the graph of each function with the graphs of its first and second derivatives.

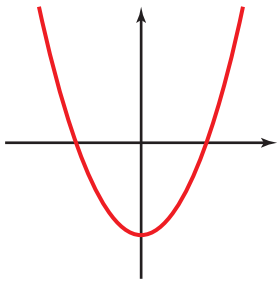
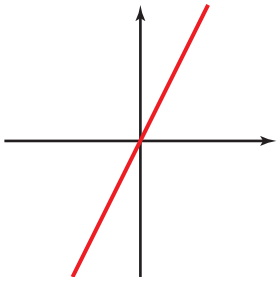
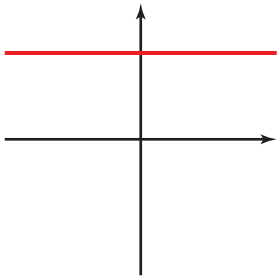
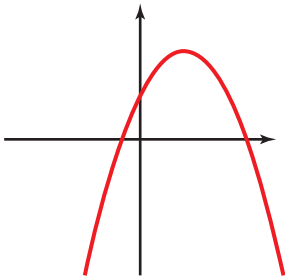
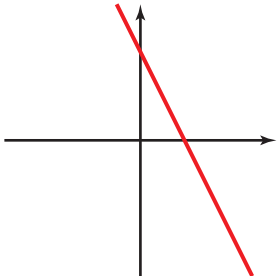
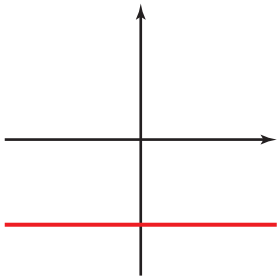
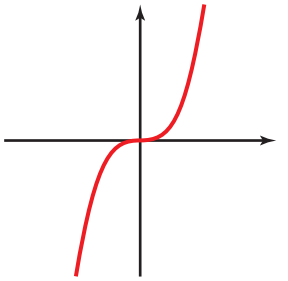
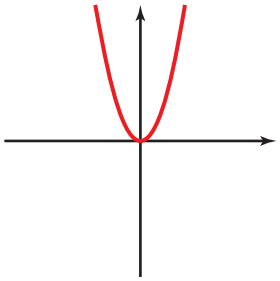
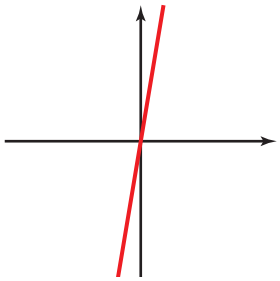
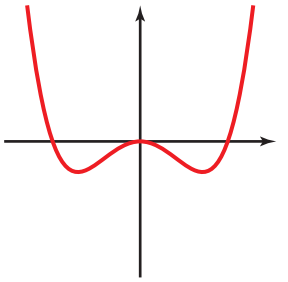
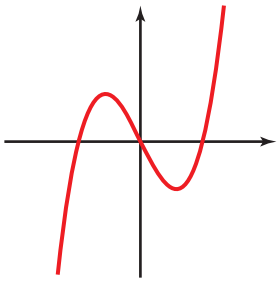
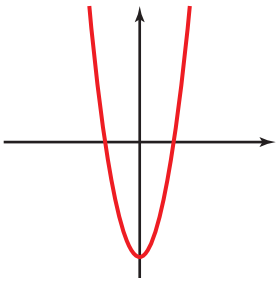
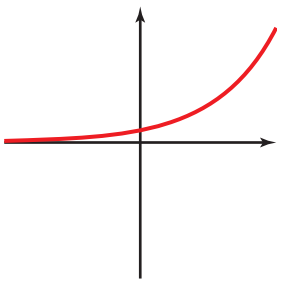
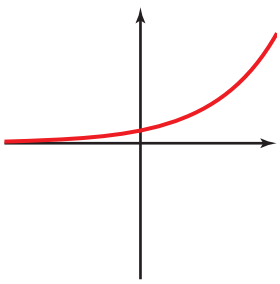
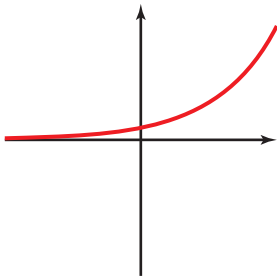


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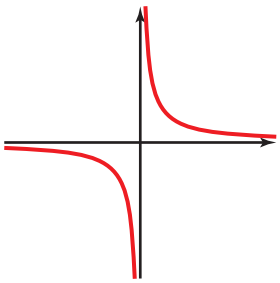
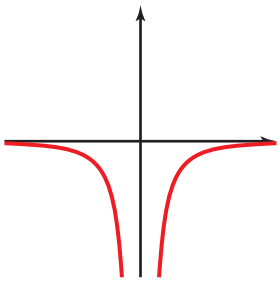
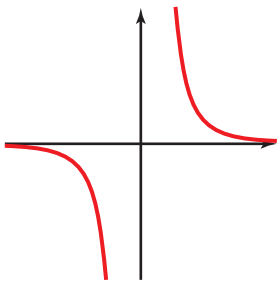
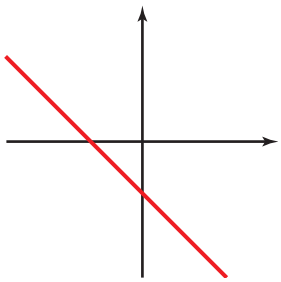
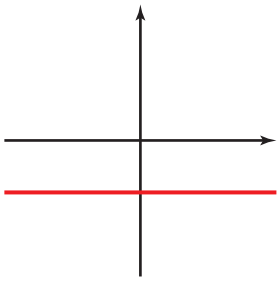
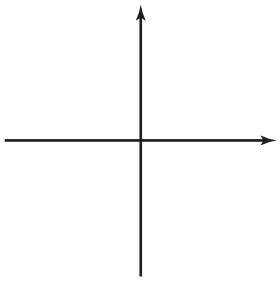
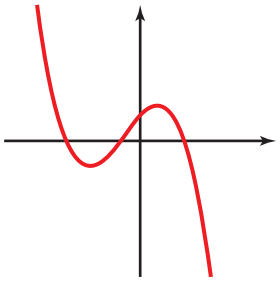
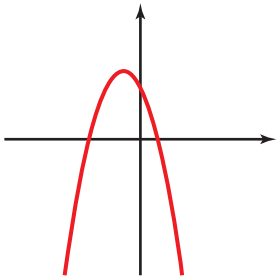
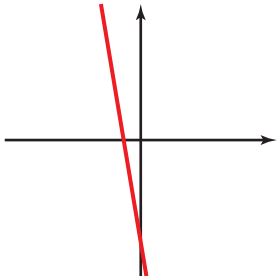
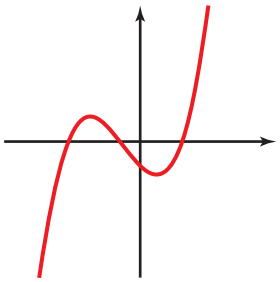
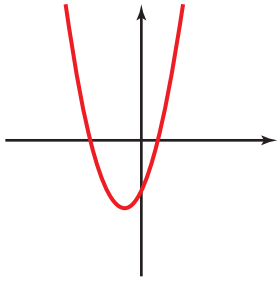
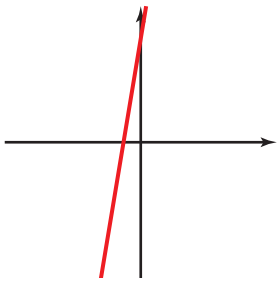
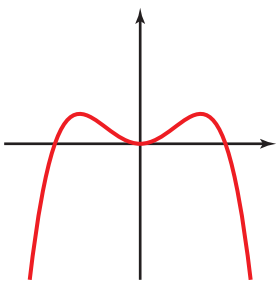
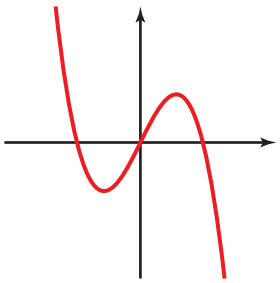
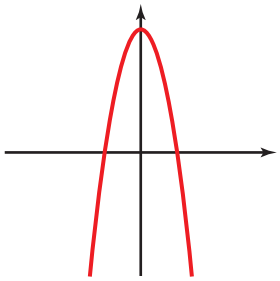
$f_5(x)$ 	$g_5(x)$ 	$h_5(x)$ 
$f_6(x)$ 	$g_6(x)$ 	$h_6(x)$ 
$f_7(x)$ 	$g_7(x)$ 	$h_7(x)$ 
$f_8(x)$ 	$g_8(x)$ 	$h_8(x)$ 
$f_9(x)$ 	$g_9(x)$ 	$h_9(x)$ 
$f_{10}(x)$ 	$g_{10}(x)$ 	$h_{10}(x)$ 

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Answers

$f_{10}(x)$ 	$g_1(x)$ 	$h_7(x)$ 
$f_6(x)$ 	$g_5(x)$ 	$h_3(x)$ 
$f_3(x)$ 	$g_4(x)$ 	$h_9(x)$ 
$f_7(x)$ 	$g_8(x)$ 	$h_{10}(x)$ 
$f_2(x)$ 	$g_9(x)$ 	$h_4(x)$ 

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$f_9(x)$ 	$g_7(x)$ 	$h_6(x)$ 
$f_8(x)$ 	$g_6(x)$ 	$h_1(x)$ 
$f_5(x)$ 	$g_2(x)$ 	$h_8(x)$ 
$f_4(x)$ 	$g_{10}(x)$ 	$h_5(x)$ 
$f_1(x)$ 	$g_3(x)$ 	$h_2(x)$ 

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